PRISTINI
SCHOOL OF AI

Chapter 2 Use Case

Dr. Ameni Youssfi Nouira

ISDesign 2026/2027

Outline

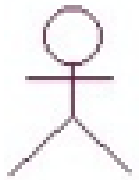
- **Presentation of the Use Case Diagram**
  - Basic Concepts
  - Refinement Concepts
  - Use Case Documentation
  - Activities

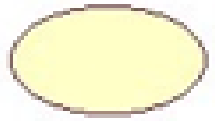
Presentation

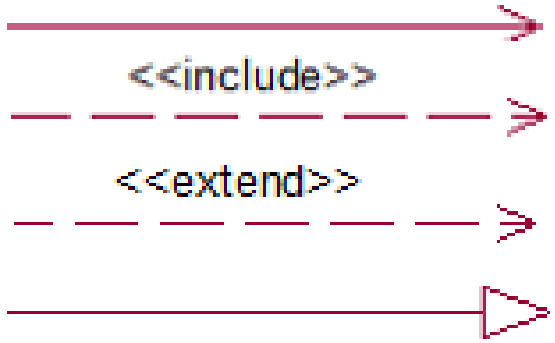
- **Use Case Diagram:**
  - The functional UML diagram
  - A way to specify the uses/features of a system
  - Represents the interactions between users and the system
  - A graphical representation (diagram) accompanied by a textual description

Basics concepts

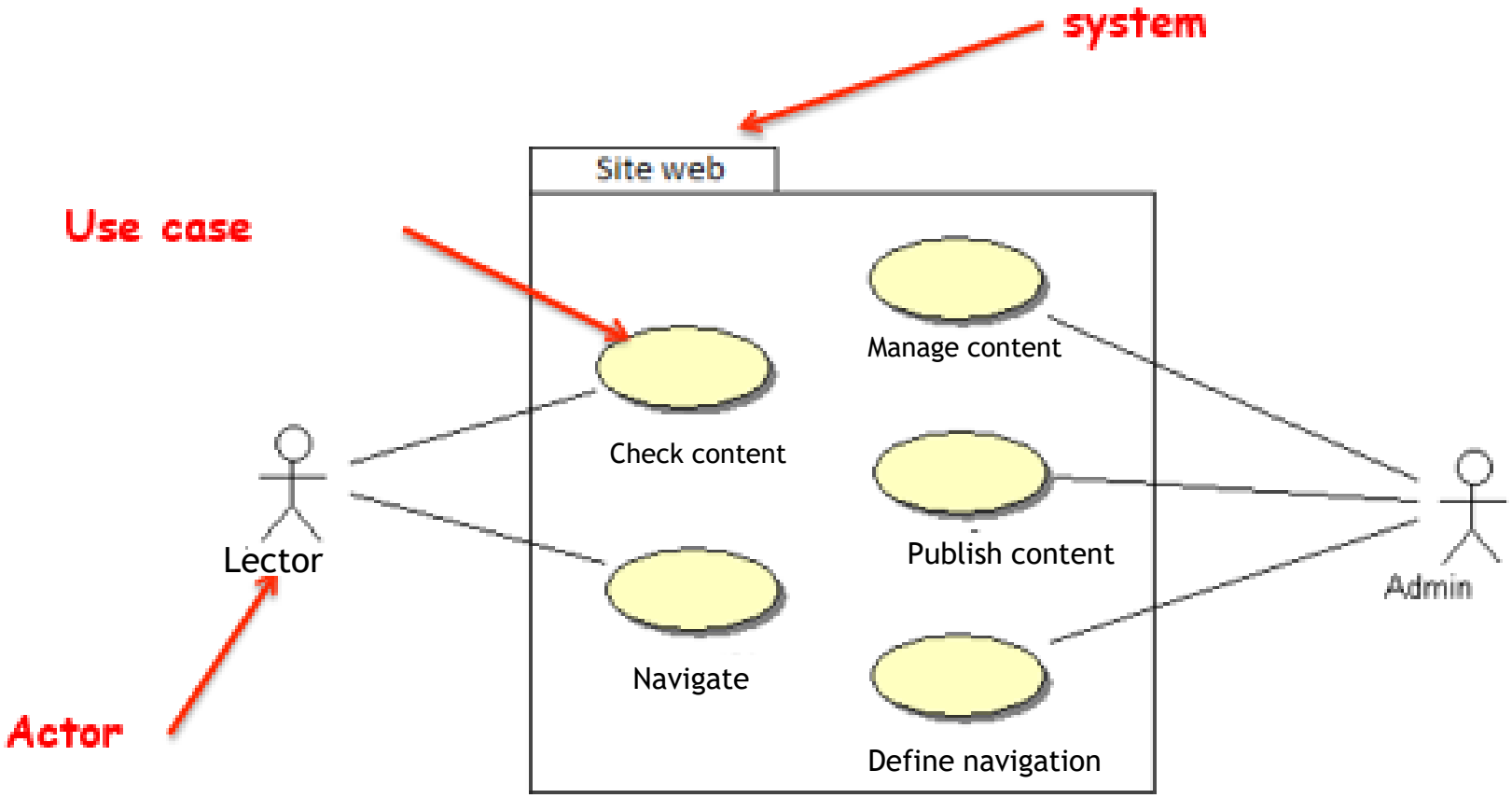
- **Actor**
  - Role played by any external entity that interacts with the system
- **Use Case**
  - Specification of a set of actions performed by the system
- **Relationships:**
  - Interaction
  - Include
  - Extend
  - Generalization







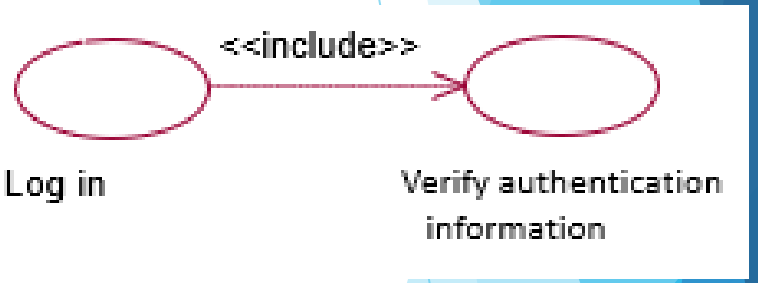
Basics


system
Site web
Use case
Manage content
Check content
Publish content
Lector
Admin
Navigate
Define navigation
Actor

Basics

- **Interaction:**
  - Relationship between an actor and a use case (UC)
- **Include:**
  - A UC contains the actions defined in another UC
- **Extend:**
  - A UC can be extended by another UC

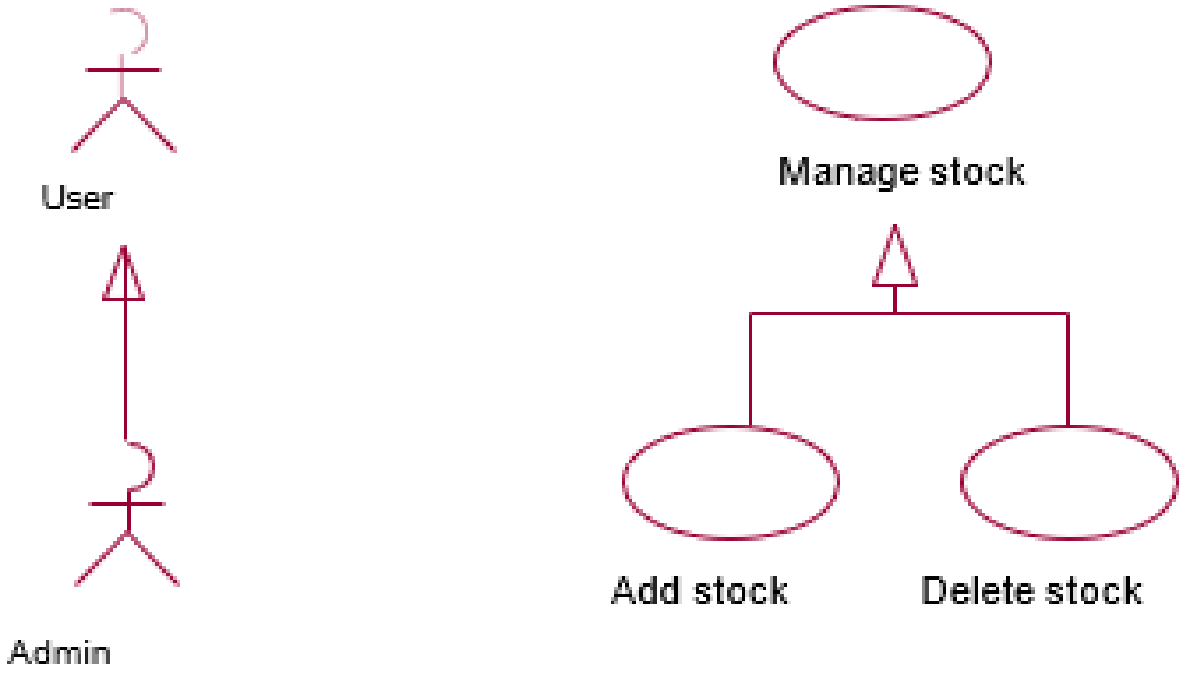






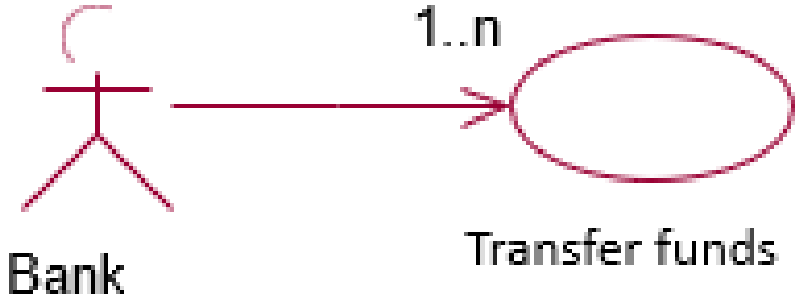
Generalization:

- A UC is a specification of another UC
- An actor is a specification of another actor



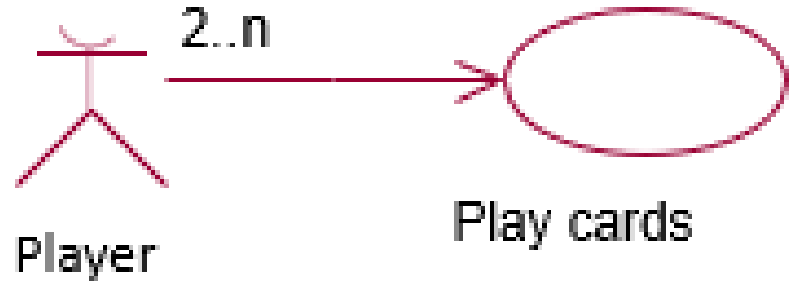
Refinement Concepts (1)

- **Cardinalities:**
  - An actor is connected to a UC, with a multiplicity >1 located at the end of the UC
  - Actor involved in multiple UCs of this type
  - Nature of involvement (not defined in the UML specification):
    - In parallel
    - Mutual exclusion
    - At different points in time



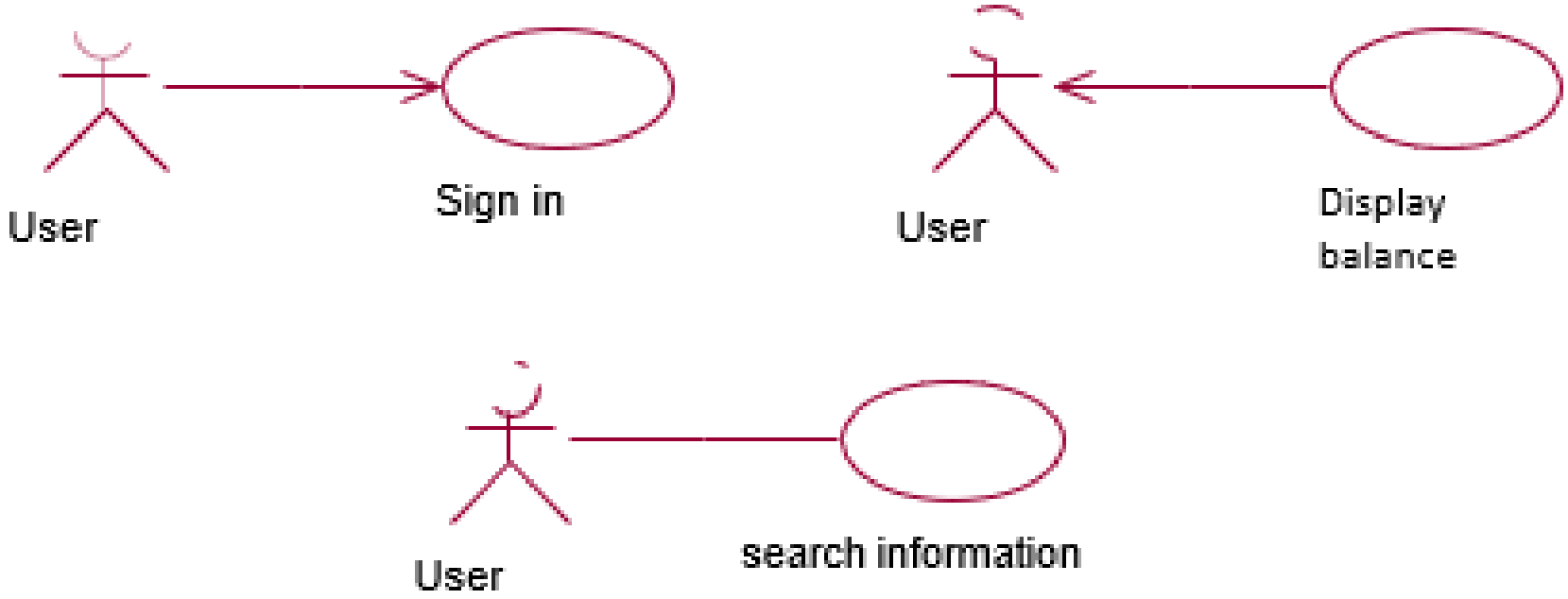
Refinement Concepts (2)

- **Cardinalities (continued):**
- A UC is connected to an actor, with a multiplicity >1 located at the end of the actor
- Multiple instances of this actor are involved in performing this UC
- The way of involvement depends on the current situation:
  - Simultaneous (concurrent) actions by different actors
  - Complementary and successive actions by the actors



Refinement Concepts (3)

- **Interaction Orientation:**
- In general, represents the direction of the interaction
- Absence of orientation: bidirectional



Use Case Documentation

- **Use Case Name:**
- **Actors involved:**
- **Description:** // brief description of the UC
- **Preconditions:** // to be verified before the UC begins
- **Main flow / Basic sequence:**
  - …
  - …
- **Alternative flows / Exceptions:**
  If… then
  If… then
- **Postconditions:** // to be verified after the UC ends

Activity

- A **library** wants a system to manage books. The system should allow:
- A **Member** to:
  - Search for a book
  - Borrow a book
  - Return a book
- A **Librarian** to:
  - Add a new book to the catalog
  - Remove a book
  - Register a new member

Task: Draw the **Use Case Diagram** with the main actors and use cases.